FED. SUP CLASS
5307

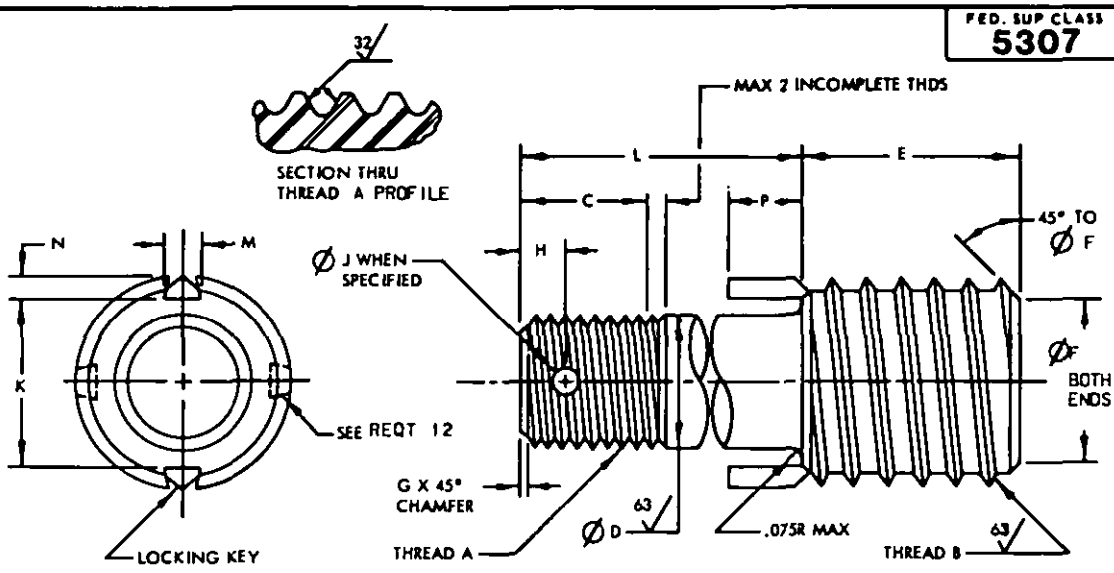



TABLE I (COARSE THREAD NUT END)

| DASH NO. | A NUT END THREAD UNJC-3A | B STUD END THREAD MOD MINOR DIA | | C* | ØD (GRIP DIA) | E | ØF | G | H | ØJ | K | KEY DIMENSIONS | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SIZE | SIZE | MINOR DIA | ± .030 | | ± .015 | ± .010 | | ± .020 | ± .010 | ± .004 | M ± .002 | N ± .004 | P ± .015 |
| -101 | .1900-24 | .3125-18UNC-2A | .256<br>.249 | .410 | .189<br>.185 | .312 | .250 | .060<br>.030 | .141 | .070 | .220 | .068 | .040 | .160 |
| -102 | .2500-20 | .3750-16UNC-2A | .320<br>.312 | .470 | .249<br>.245 | .375 | .310 | .060<br>.030 | .156 | .076 | .284 | .068 | .040 | .190 |
| -103 | .3125-18 | .4375-14UNC-2A | .383<br>.375 | .530 | .312<br>.308 | .437 | .375 | .060<br>.030 | .172 | .076 | .345 | .068 | .040 | .190 |
| -104 | .3750-16 | .5000-13UNC-2A | .440<br>.432 | .640 | .374<br>.370 | .500 | .430 | .060<br>.030 | .172 | .106 | .407 | .068 | .040 | .190 |
| -105 | .4375-14 | .5625-12UNC-2A | .503<br>.495 | .660 | .437<br>.433 | .562 | .495 | .060<br>.030 | .172 | .106 | .469 | .068 | .040 | .190 |
| -106 | .5000-13 | .6250-11UNC-2A | .565<br>.551 | .780 | .499<br>.495 | .625 | .550 | .060<br>.030 | .172 | .106 | .532 | .068 | .040 | .190 |

*THIS DIMENSION DOES NOT APPLY TO DASH NUMBERS IN TABLE III ABOVE THE HEAVY LINES.

(D) ENTIRE STANDARD REVISED

APPROVED 1 DEC 1967 REVISED (A) 25 NOV 1970 (B) 21 APRIL 1980 (C) 27 OCT 1980 (D) 7 DEC 84

User activities:
ARMY-ER, AT, ME
NAVY-MC

Review activities:
ARMY-MI, AV, GL
NAVY-SH
AIR FORCE - 82, 11
NSA-NS
DLA-IS

This Military Standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

| P.A. | AR |
|---|---|
| Other Cust | AS |
| | 99 |

TITLE: STUD, LOCKED IN — KEY LOCKED, LIGHTWEIGHT

MILITARY STANDARD
MS 51833

PROCUREMENT SPECIFICATION: MIL-S-45915

SUPERSEDES:

PAGE 1 OF 4

DD FORM 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

5307-0336

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User activities
ARMY - ER, AT, ME
NAVY - MC

Review activities
ARMY - MI, AV, GL
NAVY - SH
AIR FORCE - 82, 11
NSA - NS
DLA - IS

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TABLE II (FINE THREAD NUT END)

| DASH NO. | A NUT END THREAD UNJF-3A SIZE | B STUD END THREAD MOD MINOR DIA SIZE | B MINOR DIA | C ± .030 | ∅ D (GRIP DIA) | E ± .015 | ∅ F ± .010 | G | H ± .020 | ∅ J ± .010 | K ± .004 | KEY DIMENSIONS M ± .002 | N ± .004 | P ± .015 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -201 | .1900-32 | .3125-18UNC-2A | .256<br>.249 | .410 | .189<br>.185 | .312 | .250 | .060<br>.030 | .141 | .070 | .220 | .068 | .040 | .160 |
| -202 | .2500-28 | .3750-16UNC-2A | .320<br>.312 | .470 | .249<br>.245 | .375 | .310 | .060<br>.030 | .156 | .076 | .284 | .068 | .040 | .190 |
| -203 | .3125-24 | .4375-14UNC-2A | .383<br>.375 | .530 | .312<br>.308 | .437 | .375 | .060<br>.030 | .172 | .076 | .345 | .068 | .040 | .190 |
| -204 | .3750-24 | .5000-13UNC-2A | .440<br>.432 | .640 | .374<br>.370 | .500 | .430 | .060<br>.030 | .172 | .106 | .407 | .068 | .040 | .190 |
| -205 | .4375-20 | .5625-12UNC-2A | .500<br>.495 | .660 | .437<br>.433 | .562 | .495 | .060<br>.030 | .172 | .106 | .469 | .068 | .040 | .190 |
| -206 | .5000-20 | .6250-11UNC-2A | .565<br>.551 | .700 | .499<br>.495 | .625 | .550 | .060<br>.030 | .172 | .106 | .532 | .068 | .040 | .190 |

REQUIREMENTS:

1. MATERIAL: STEEL, CORROSION-RESISTANT, TYPE 303(UNS S30300) IN ACCORDANCE WITH AMS 5640M-79 OR ASTM A582-60.
   STEEL, ALLOY, GRADE 4140(UNS G41400) IN ACCORDANCE WITH MIL-S-5626.
   STEEL, CORROSION-RESISTANT, TYPE A286 (UNS S66286) IN ACCORDANCE WITH AMS5734D-76.
   LOCKING KEYS: STEEL, CORROSION-RESISTANT, TYPE 302 (UNS S30200) IN ACCORDANCE WITH QQ-W-423.
2. PROTECTIVE COATING OR TREATMENT: ALLOY STEEL SHALL BE CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3. CORROSION-RESISTANT STEEL SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH ASTM A380-78.
3. SURFACE ROUGHNESS: MACHINED SURFACES SHALL BE IN ACCORDANCE WITH ANSI B46.1-78. UNLESS OTHERWISE SPECIFIED, MACHINED SURFACES SHALL BE 125 MICROINCHES.
4. THREADS: NUT END THREADS SHALL CONFORM TO MIL-S-8879. STUD END THREADS SHALL CONFORM TO FED-STD-H28/2 EXCEPT MINOR DIAMETERS SHALL BE MODIFIED AS SPECIFIED HEREIN.
5. HEAT TREATMENT: ALLOY STEEL STUDS, 4140 SHALL BE HEAT TREATED TO 160,000 PSI $F_{tu}$ MINIMUM IN ACCORDANCE WITH MIL-H-6875. CORROSION-RESISTANT STEEL STUDS, A286 SHALL BE HEAT TREATED TO 140,000 PSI $F_{tu}$ MINIMUM.
6. HARDNESS: ALLOY STEEL 4140 SHALL HAVE A HARDNESS RANGE OF 36.40 HRC.

APPROVED 1 DEC 1967 REVISED (D) FOR CHANGES SEE PAGES 1, 2, 3 AND 4

| P.A. | AR | INTERNATIONAL INTEREST | TITLE | MILITARY STANDARD |
|---|---|---|---|---|
| Other Cust | AS<br>99 | | STUD, LOCKED IN-KEY LOCKED, LIGHTWEIGHT | MS 51833 |
| PROCUREMENT SPECIFICATION: MIL-S-45915 | | SUPERSEDES: | | PAGE 2 OF 4 |



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7. PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS NUMBER PLUS THE DASH NUMBERS AND THE FOLLOWING AS APPLICABLE:

EXAMPLE: MS51833 - 103 D 24

- MS51833: STUD, LOCKED IN-KEY LOCKED
- 24: 1.500 LENGTH OF NUT END (SEE TABLE III)
- D: DRILLED HOLE IN NUT END. USE "DASH" IN LIEU OF "D" IF DRILLED HOLE NOT REQUIRED.
- 103: .3125-18 NUT END THREAD - .4375-14 STUD END THREAD
- MATERIAL: ALLOY STEEL, 4140
  CRES, A286, ADD "A" IN LIEU OF DASH
  CRES, 303, ADD "C" IN LIEU OF DASH

8. STUDS SHALL BE FREE FROM ALL HANGING BURRS AND SLIVERS WHICH MIGHT BECOME DISLODGED UNDER USAGE.
9. ALL DIMENSIONS ARE AFTER PROTECTIVE COATING OR TREATMENT.
10. TOLERANCE ON ANGLES IS ±2°.
11. *GRIP DIAMETER SHALL BE CONCENTRIC TO PITCH DIAMETER OF STUD END THREAD WITHIN .006 FIM.*
12. STUDS WITH NUT END THREAD SIZE .1900 OR .2500 SHALL BE SUPPLIED WITH TWO (2) LOCKING KEYS SPACED 180° APART. ALL OTHER SIZES SHALL BE SUPPLIED WITH FOUR (4) LOCKING KEYS SPACED 90° APART.
13. MARKING:
    CRES 303 HAS NO IDENTIFYING MARK.
    ALLOY STEEL 4140 SHALL BE IDENTIFIED ON TOP OF STUD BY TWO (2) PARALLEL DASH MARKS.
    CRES A286 SHALL BE IDENTIFIED ON TOP OF STUD BY ONE DASH MARK.

NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. INSTALLATION OF STUD SHALL BE IN ACCORDANCE WITH MS51835.
3. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
4. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS, OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE *ADOPTED*.

APPROVED 1 DEC 1967 REVISED (D) FOR CHANGES SEE PAGES 1, 2, 3 AND 4

User activities:
ARMY - ER, AT, ME
NAVY - MC

Review activities
ARMY -MI, AV, GL
NAVY -SH
AIR FORCE - 82, 11
NSA - NS
DLA - IS

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| P. A. AR | INTERNATIONAL INTEREST | TITLE | MILITARY STANDARD |
|---|---|---|---|
| Other Cust AS 99 | | STUD LOCKED IN-KEY LOCKED LIGHTWEIGHT | MS 51833 |
| PROCUREMENT SPECIFICATION MIL-S-45915 | SUPERSEDES | | PAGE 3 OF 4 |



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Review activities:
ARMY - MI, AV, GL
NAVY - SH
AIR FORCE - 82, 11
NSA - NS
DLA - IS

User activities
ARMY - ER, AT, ME
NAVY - MC



TABLE III - TABULATED LENGTHS

| L NUT END ± .030 | LENGTH DASH NUMBER NOMINAL DIAMETER .1900 | .2500 | .3125 | .3750 | .4375 | .5000 |
|---|---|---|---|---|---|---|
| .375 | -6 | -6 | | | | |
| .438 | -7 | -7 | | | | |
| .500 | -8 | -8 | | | | |
| .562 | -9 | -9 | -9 | | | |
| .625 | -10 | -10 | -10 | -10 | -10 | |
| .688 | -11 | -11 | -11 | -11 | -11 | |
| .750 | -12 | -12 | -12 | -12 | -12 | -12 |
| .812 | -13 | -13 | -13 | -13 | -13 | -13 |
| .875 | -14 | -14 | -14 | -14 | -14 | -14 |
| .938 | -15 | -15 | -15 | -15 | -15 | -15 |
| 1.000 | -16 | -16 | -16 | -16 | -16 | -16 |
| 1.062 | -17 | -17 | -17 | -17 | -17 | -17 |
| 1.125 | -18 | -18 | -18 | -18 | -18 | -18 |
| 1.188 | -19 | -19 | -19 | -19 | -19 | -19 |
| 1.250 | -20 | -20 | -20 | -20 | -20 | -20 |
| 1.312 | -21 | -21 | -21 | -21 | -21 | -21 |
| 1.375 | -22 | -22 | -22 | -22 | -22 | -22 |
| 1.438 | -23 | -23 | -23 | -23 | -23 | -23 |
| 1.500 | -24 | -24 | -24 | -24 | -24 | -24 |
| 1.562 | -25 | -25 | -25 | -25 | -25 | -25 |
| 1.625 | -26 | -26 | -26 | -26 | -26 | -26 |
| 1.688 | -27 | -27 | -27 | -27 | -27 | -27 |
| 1.750 | -28 | -28 | -28 | -28 | -28 | -28 |
| 1.812 | -29 | -29 | -29 | -29 | -29 | -29 |
| 1.875 | -30 | -30 | -30 | -30 | -30 | -30 |
| 1.938 | -31 | -31 | -31 | -31 | -31 | -31 |
| 2.000 | -32 | -32 | -32 | -32 | -32 | -32 |
| 2.125 | -34 | -34 | -34 | -34 | -34 | -34 |
| 2.250 | -36 | -36 | -36 | -36 | -36 | -36 |
| 2.375 | -38 | -38 | -38 | -38 | -38 | -38 |
| 2.500 | -40 | -40 | -40 | -40 | -40 | -40 |
| 2.625 | | -42 | -42 | -42 | -42 | -42 |
| 2.750 | | -44 | -44 | -44 | -44 | -44 |
| 2.875 | | -46 | -46 | -46 | -46 | -46 |
| 3.000 | | -48 | -48 | -48 | -48 | -48 |
| 3.125 | | -50 | -50 | -50 | | -50 |
| 3.250 | | -52 | -52 | -52 | | -52 |
| 3.375 | | -54 | -54 | -54 | | -54 |
| 3.500 | | -56 | -56 | -56 | | -56 |
| 3.625 | | -58 | -58 | -58 | | -58 |
| 3.750 | | -60 | -60 | -60 | | -60 |
| 3.875 | | -62 | -62 | -62 | | -62 |
| 4.000 | | -64 | -64 | -64 | | -64 |

DASH NUMBERED PARTS ABOVE THE HEAVY LINE ARE THREADED TO THE STUD END AS SPECIFIED HEREIN. THE MAXIMUM DISTANCE FROM THE STUD END TO THE FIRST FULL THREAD SHALL BE EQUAL TO THE SUM OF THE MAXIMUM FILLET RADIUS AND A MAXIMUM OF TWO INCOMPLETE THREADS. INCOMPLETED THREADS NOT TO ENTER FILLET AREA.

APPROVED 1 DEC 1967 REVISED (D) FOR CHANGES SEE PAGES 1, 2, 3, AND 4

| P. A. AR<br>Other Cust AS<br>99 | INTERNATIONAL INTEREST | TITLE<br>STUD, LOCKED IN - KEY LOCKED, LIGHTWEIGHT | MILITARY STANDARD<br>MS 51833 |
|---|---|---|---|
| PROCUREMENT SPECIFICATION<br>MIL-S-45915 | SUPERSEDES | | PAGE 4 OF 4 |



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